Supplementary Material

5.5. Defined Reactions and Corresponding Rate Constants

The reactions defined in our model are listed together with their corresponding k_i rate constants in Table 5.1. The sources for the reactions are the same as they are for the corresponding rate constants. The right side of the first reaction (k_1) corresponds to cells of the well that are not modeled but in an experiment would still be consuming glucose and thus influence the concentrations of GLUCOUT, which corresponds to glucose in the whole cell culture.

Table 5.1: The reactions of the model and the original and calibrated rate constants used in the model are provided in this table. The values are given using scientific notation (i.e., E-08 represents 10^{-8}) in units of $\frac{1}{s}$ for first order rate constants and in units of $\frac{cm^3}{mol \cdot s}$ for rates of second order reactions. The original value corresponds to the one derived from the literature, while the calibrated value represents the revised constants that we used in our simulations. Revision of rate constants was guided by ensuring the stability of the basal state statistical calculations (refer to Section 3.2.3.2) as well as by trial and error to represent biologically realistic qualitative dynamics.

k_i	Reaction	Original	Calibrated	Source
k_1	GLUCOUT $\longrightarrow \varnothing$	2.5E-08	2.5E-08	calibrated
k_2	$GLUT_1$ + GLUCOUT $\longrightarrow$ $GLUT_1$_GLUCOUT	1.28E+06	1.28E+06	[91]
k_3	$GLUT_1$_GLUCOUT $\longrightarrow$ $GLUT_1$ + GLUCOUT	1.553	0.555	[91]
k_4	$GLUT_1$_GLUCOUT $\longrightarrow$ $GLUT_1$ + GLUCIN	0.111	0.111	[91]
k_5	$GLUT_1$ + GLUCIN $\longrightarrow$ $GLUT_1$_GLUCIN	1.28E+06	1.28E+06	[91]
k_6	$GLUT_1$_GLUCIN $\longrightarrow$ $GLUT_1$ + GLUCIN	1.553	0.555	[91]
k_7	$GLUT_1$_GLUCIN $\longrightarrow$ $GLUT_1$ + GLUCOUT	0.111	0.111	[91]
k_8	GLUCIN $\longrightarrow$ 2 PYRUVATE + 7 Q	4E-05	4E-05	calibrated
k_9	PYRUVATE $\longrightarrow$ 12.5 Q	4E-05	4E-05	calibrated
k_{10}	Q + ADP $\longrightarrow$ ATP	1.0E+06	1.0E+06	calibrated
k_{11}	ATP $\longrightarrow$ ADP	1.3E-04	6.15E-05	[91]
k_{12}	2 ADP $\longrightarrow$ ATP + AMP	1.67E+07	1.67E+05	[91]
k_{13}	ATP + AMP $\longrightarrow$ 2 ADP	1.67E+07	1.67E+04	[91]
k_{14}	ATP + AMPK $\longrightarrow$ AMPK_ATP	1E+06	1E+06	[92]
k_{15}	AMPK_ATP $\longrightarrow$ ATP + AMPK	2.2E-01	2.2E-01	[92]
k_{16}	ADP + AMPK $\longrightarrow$ AMPK_ADP	1.5E+06	1.5E+06	[92]
k_{17}	AMPK_ADP $\longrightarrow$ ADP + AMPK	2.25E-03	2.25E-03	[92]
k_{18}	AMP + AMPK $\longrightarrow$ AMPK_AMP	1.5E+06	1.5E+06	[92]
k_{19}	AMPK_AMP $\longrightarrow$ AMP + AMPK	3.75E-03	3.75E-03	[92]
k_{20}	ATP + AMPKA $\longrightarrow$ AMPKA_ATP	1E+06	1E+06	[92]
k_{21}	AMPKA_ATP $\longrightarrow$ ATP + AMPKA	4E-01	4E-01	[92]
k_{22}	ADP + AMPKA $\longrightarrow$ AMPKA_ADP	1.5E+06	1.5E+06	[92]
k_{23}	AMPKA_ADP $\longrightarrow$ ADP + AMPKA	2.25E-03	2.25E-03	[92]
k_{24}	AMP + AMPKA $\longrightarrow$ AMPKA_AMP	1E+06	1E+06	[92]
k_{25}	AMPKA_AMP $\longrightarrow$ AMP + AMPKA	3.75E-03	3.75E-03	[92]
k_{26}	AMPK + LKB_1 $\longrightarrow$ AMPK_LKB_1	3.61E+06	3.61E+06	[92]
k_{27}	AMPK_LKB_1 $\longrightarrow$ AMPK + LKB_1	5E+03	5E+03	[92]
k_{28}	AMPK_LKB_1 $\longrightarrow$ AMPKA + LKB_1	5E+01	5E+01	[92]
k_{29}	AMPKA + PP_2A $\longrightarrow$ AMPKA_PP_2A	8.21E+06	8.21E+06	[92]
k_{30}	AMPKA_PP_2A $\longrightarrow$ AMPKA + PP_2A	5E+02	5E+02	[92]
k_{31}	AMPKA_PP_2A $\longrightarrow$ AMPK + PP_2A	5E+01	5E+01	[92]
k_{32}	AMPK_ATP + LKB_1 $\longrightarrow$ AMPK_ATP_LKB_1	3.61E+06	3.61E+06	[92]
k_{33}	AMPK_ATP_LKB_1 $\longrightarrow$ AMPK_ATP + LKB_1	5E+03	5E+03	[92]
k_{34}	AMPK_ATP_LKB_1 $\longrightarrow$ AMPKA_ATP + LKB_1	5E+01	5E+00	[92]

k_i	Reaction	Original	Calibrated	Source
k_{35}	AMPKA_ATP + PP_2A ⟶ AMPKA_ATP_PP_2A	8.21E+06	8.21E+06	[92]
k_{36}	AMPKA_ATP_PP_2A ⟶ AMPKA_ATP + PP_2A	5E+02	5E+02	[92]
k_{37}	AMPKA_ATP_PP_2A ⟶ AMPK_ATP + PP_2A	5E+01	5E+02	[92]
k_{38}	AMPK_ADP + LKB_1 ⟶ AMPK_ADP_LKB_1	5.41E+06	5.41E+06	[92]
k_{39}	AMPK_ADP_LKB_1 ⟶ AMPK_ADP + LKB_1	7.5E+03	7.5E+01	[92]
k_{40}	AMPK_ADP_LKB_1 ⟶ AMPKA_ADP + LKB_1	7.5E+01	1.5E+02	[92]
k_{41}	AMPKA_ADP + PP_2A ⟶ AMPKA_ADP_PP_2A	8.21E+04	8.21E+04	[92]
k_{42}	AMPKA_ADP_PP_2A ⟶ AMPKA_ADP + PP_2A	5E+00	5E+00	[92]
k_{43}	AMPKA_ADP_PP_2A ⟶ AMPK_ADP + PP_2A	1E+00	1E+00	[92]
k_{44}	AMPK_AMP + LKB_1 ⟶ AMPK_AMP_LKB_1	1.44E+07	1.44E+07	[92]
k_{45}	AMPK_AMP_LKB_1 ⟶ AMPK_AMP + LKB_1	2E+04	2E+01	[92]
k_{46}	AMPK_AMP_LKB_1 ⟶ AMPKA_AMP + LKB_1	2E+02	3E+02	[92]
k_{47}	AMPKA_AMP + PP_2A ⟶ AMPKA_AMP_PP_2A	8.21E+04	8.21E+04	[92]
k_{48}	AMPKA_AMP_PP_2A ⟶ AMPKA_AMP + PP_2A	5E+00	5E+00	[92]
k_{49}	AMPKA_AMP_PP_2A ⟶ AMPK_AMP + PP_2A	1E+00	1E+00	[92]
k_{50}	AMPKA + MTORA ⟶ AMPKA + MTOR	8.01423E+07	8.01423E+07	[81]
k_{51}	AMPKA_ATP + MTORA ⟶ AMPKA_ATP + MTOR	8.01423E+07	8.01423E+07	[81]
k_{52}	AMPKA_ADP + MTORA ⟶ AMPKA_ADP + MTOR	8.01423E+07	8.01423E+07	[81]
k_{53}	AMPKA_AMP + MTORA ⟶ AMPKA_AMP + MTOR	4.54269E+08	4.54269E+08	[81]
k_{54}	MTORA + AMPKA ⟶ MTORA + AMPK	9.6385E+03	9.6385E+03	[81]
k_{55}	MTORA + AMPKA_ATP ⟶ MTORA + AMPK_ATP	9.6385E+03	9.6385E+03	[81]
k_{56}	MTORA + AMPKA_ADP ⟶ MTORA + AMPK_ADP	9.6385E+03	9.6385E+03	[81]
k_{57}	MTORA + AMPKA_AMP ⟶ MTORA + AMPK_AMP	9.6385E+03	9.6385E+03	[81]
k_{58}	ULKA + MTORA ⟶ MTORA + ULK	1.6122E+04	1.6122E+04	[81]
k_{59}	ULKA + MTORA ⟶ MTOR + ULKA	1.9507E+05	1.9507E+05	[81]
k_{60}	AMPKA + TSC ⟶ AMPKA + TSCA	1.489E+03	1.489E+03	[81]
k_{61}	AMPKA_ATP + TSC ⟶ AMPKA_ATP + TSCA	1.489E+03	1.489E+03	[81]
k_{62}	AMPKA_ADP + TSC ⟶ AMPKA_ADP + TSCA	1.5342E+03	1.5342E+03	[81]
k_{63}	AMPKA_AMP + TSC ⟶ AMPKA_AMP + TSCA	1.3537E+04	1.3537E+04	[81]
k_{64}	RHEBA + TSCA ⟶ RHEB + TSCA	5.2292E+03	5.2292E+03	[81]
k_{65}	RHEBA + MTOR ⟶ RHEBA + MTORA	1.4312E+08	1.4312E+08	[81]
k_{66}	RHEB ⟶ RHEBA	9.2104E-08	9.2104E-08	[81]
k_{67}	AKTA ⟶ AKT	5.0667E-07	5.0667E-07	[81]
k_{68}	AKTA + TSCA ⟶ AKTA + TSC	6.4680E+09	6.4680E+09	[81]
k_{69}	AMPKA + ULK ⟶ AMPKA + ULKA	4.0270E+03	3.3223E+05	[81]
k_{70}	AMPKA_ATP + ULK ⟶ AMPKA_ATP + ULKA	4.0270E+03	3.3223E+05	[81]
k_{71}	AMPKA_ADP + ULK ⟶ AMPKA_ADP + ULKA	4.0270E+03	3.423E+05	[81]
k_{72}	AMPKA_AMP + ULK ⟶ AMPKA_AMP + ULKA	4.0270E+03	1.2203E+06	[81]
k_{73}	ULKA ⟶ ULK	2.8119E-05	2.8119E-05	[81]
k_{74}	AMPKA + ULKA ⟶ AMPK + ULKA	2.5661E+04	4.1653E+01	[81]
k_{75}	AMPKA_ATP + ULKA ⟶ AMPK_ATP + ULKA	2.6661E+04	4.1653E+01	[81]
k_{76}	AMPKA_ADP + ULKA ⟶ AMPK_ADP + ULKA	2.6661E+04	4.1653E+01	[81]
k_{77}	AMPKA_AMP + ULKA ⟶ AMPK_AMP + ULKA	2.6661E+04	4.1653E+01	[81]

5.6. Initial Conditions

Table 5.2: The table presents the basal state initial condition ranges for our simulations. The IC of each species for any given simulation is randomly generated around the species mean (3. column) for both basal and starvation conditions. The random sampling range for a species s is defined as follows: $[0.5 \cdot \bar{x}_s, 1.5 \cdot \bar{x}_s]$, where $\bar{x}_s$ denotes the species' mean concentration. In columns min and max, the lower and upper bounds of the literature-derived concentration ranges are provided. The units for each concentration value are $\frac{mol}{cm^3}$. An IC of 0 $[\frac{mol}{cm^3}]$ is assumed for all enzyme-substrate Michaelis-Menten complexes (bottom 10 rows). Initial Conditions for GLUCOUT depend on the experimental conditions and is an input of the system: for starvation the concentration is set at $1E-6$ $[\frac{mol}{cm^3}]$, while for basal-state simulations, the input is $6E-6$ $[\frac{mol}{cm^3}]$. Similarly for Q (the proton motive force), in starvation the IC $1E-9$ $[\frac{mol}{cm^3}]$, whereas in homeostatic simulations it is $3E-6$ $[\frac{mol}{cm^3}]$. Sources for each IC are also provided. The descriptor column specifies if the species was used for basal state statistics (please refer to section 3.2.3.2).

species	min $[\frac{mol}{cm^3}]$	mean $[\frac{mol}{cm^3}]$	max $[\frac{mol}{cm^3}]$	source	descriptor
GLUT1	1.47E-10	1.99E-10	2.57E-10	[132]	yes
GLUCIN	8.50E-08	4.00E-07	5.50E-07	[133]	yes
ATP	3.00E-06	5.00E-06	7.00E-06	[48–50]	yes
ADP	5.00E-08	1.25E-07	2.00E-07	[92]	yes
AMP	5.00E-10	2.50E-09	5.00E-09	[92]	yes
AMPK	1.00E-10	1.00E-10	1.00E-10	[81]	yes
AMPKA	0	0	0	[81]	yes
AMPK_ATP	1.00E-10	1.00E-10	1.00E-10	[81]	yes
AMPK_ADP	1.00E-10	1.00E-10	1.00E-10	[81]	yes
AMPK_AMP	1.00E-10	1.00E-10	1.00E-10	[81]	yes
AMPKA_ATP	0.00E+00	0.00E+00	0.00E+00	[92]	yes
AMPKA_ADP	0.00E+00	0.00E+00	0.00E+00	[92]	yes
AMPKA_AMP	0.00E+00	0.00E+00	0.00E+00	[92]	yes
LKB1	2.13E-11	2.88E-11	3.72E-11	[132]	yes
PP2A	2.86E-10	3.86E-10	4.98E-10	[132]	yes
PYRUVATE	2.00E-08	3.30E-08	4.00E-08	[134]	no
AKT	0.00E+00	0.00E+00	0.00E+00	[81]	yes
AKTA	5.00E-11	1.00E-10	2.00E-10	[81]	yes
MTOR	0.00E+00	0.00E+00	0.00E+00	[81]	yes
MTORA	1.88E-10	3.75E-10	7.50E-10	[81]	yes
RHEB	0.00E+00	0.00E+00	0.00E+00	[81]	yes
RHEBA	1.50E-10	3.00E-10	6.00E-10	[81]	yes
TSC	1.13E-10	2.25E-10	4.50E-10	[81]	yes
TSCA	0.00E+00	0.00E+00	0.00E+00	[81]	yes
ULK	3.50E-11	7.00E-11	1.40E-10	[81]	yes
ULKA	0	0	0	[81]	yes
AMPK_LKB1	0	0	0	assumed	no
AMPK_ATP_LKB1	0	0	0	assumed	no
AMPK_ADP_LKB1	0	0	0	assumed	no
AMPK_AMP_LKB1	0	0	0	assumed	no
AMPKA_PP2A	0	0	0	assumed	no
AMPKA_ATP_PP2A	0	0	0	assumed	no
AMPKA_ADP_PP2A	0	0	0	assumed	no
AMPKA_AMP_PP2A	0	0	0	assumed	no
GLUT1_GLUCOUT	0	0	0	assumed	no
GLUT1_GLUCIN	0	0	0	assumed	no
input	basal	starvation		source	no
GLUCOUT	6E-6	1E-6		[45]	yes
Q	3E-6	1E-9		[45]	no